TIMES SOFTWARE IDEATHON 2026

PayShield

Catch every payroll error and fraud before salary leaves the company.

PayShield is the hero: one readiness check before every pay run. Five guards feed it: Employment, Attendance, Bank Shield, Pay Structure and Compliance.

Read the overview →

See the live demo →

Times Software Ideathon 2026 · Prepared by Varnika Raghav · 18 October 2026

All names and figures in screens are sample data.

ONE HERO PRODUCT, FIVE GUARDS

PayShield · Trust Check
Scores every record and the whole run, and sends each alert to the guard where it is fixed.

People & Employment
Is every person in this run really due this pay?

Time & Attendance
Were the paid hours really worked?

Bank Shield
Is the salary going to the right account?

Pay Structure
Is this pay right for this person, this grade and this policy?

Compliance
Are we following MOM, CPF, IRAS and our own policies?

★ Fraud Network

★ PayShield Agent checks the whole run in one prompt

OVERVIEW

PayShield on one page

PayShield is the product. The five guards are the features that make its decision trustworthy.

1

The problem

Errors and fraud are found after payday: leavers paid, salaries sent to hackers, OT above the cap, old CPF rules, unapproved raises.

See the problems →

2

PayShield Trust Check

Analyses every record and gives the run a Pay Score. It only alerts; each alert opens the guard where it is fixed. Approve unlocks at 100.

Meet PayShield →

3

Five guards feed it

Employment, Attendance, Bank Shield, Pay Structure, Compliance. Each has its own page, validators and parameters.

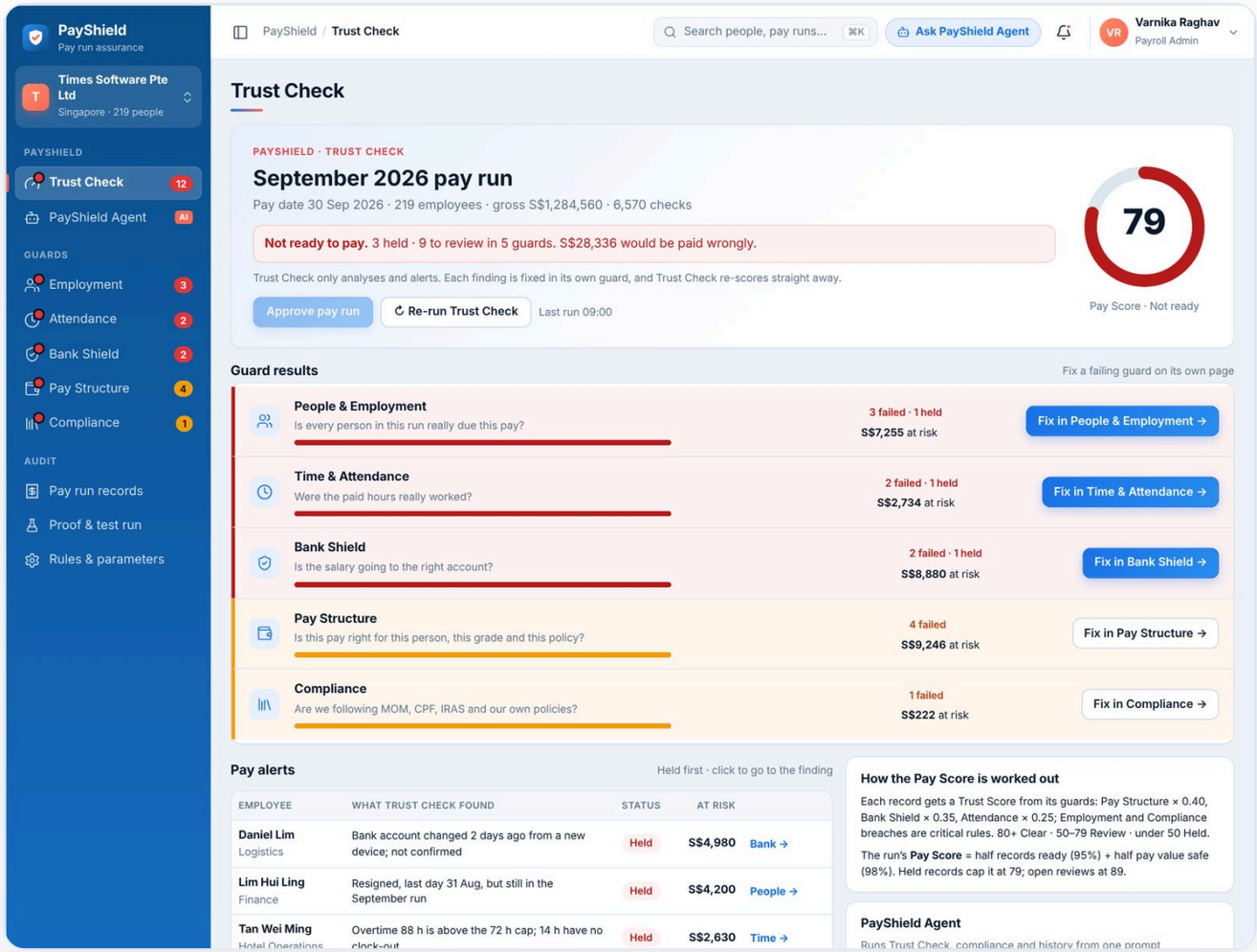
See the guards →

4

PayShield Agent, AI and proof

One prompt makes the agent check the whole run and return a Pay Score with fixes; every run is recorded. Tested: 25 of 25 hidden problems caught in each of 50 synthetic companies.

See the AI →



Trust Check in the prototype: Pay Score 79, 12 alerts across 5 guards, S$28,336 at risk before payday

RECORDS CHECKED
219
30 checks each, before approval

NEEDED A PERSON
12
instead of reviewing all 219

KEPT RIGHT
S$28,336
in one sample run

THE PROBLEM

Five problems our customers face every payday

PROBLEM	WHAT HAPPENS TODAY	WHAT IT COSTS
Errors found after payday	The pay run has no check against last month or against the rules. Mistakes surface when an employee complains.	Reruns, reprocessed pay runs, recovery of overpayments, unhappy staff
Salary sent to a hacker	A stolen login can change bank details just before payday, and nothing holds that payment.	A full salary lost; the employer often pays twice
Buddy punching and missed punches	Colleagues clock in for each other; forgotten clock-outs are chased after cutoff.	Paid hours not worked; wrong OT; HR follow-ups
Rule changes missed	When MOM, CPF or IRAS changes a rule, a developer updates code and each company finds out on its own.	Short CPF, late payment interest, compliance risk
HR time lost	Payroll teams check spreadsheets by hand and answer the same pay questions every month.	Slower pay runs; less time for real HR work

In Times Portal today, the pay run wizard goes from Configure Components straight to Review & Confirm. Errors are handled afterwards through *Rerun pay records* and *Reprocess pay run*. PayShield moves that work before the lock.

WHY

Payroll mistakes are common, costly, and the rules keep moving

Singapore figures where they exist. The one US study is marked: it shows the size of the problem, not a Singapore number.

1 in 5

payrolls contain errors

Missing or wrong time punches are the slowest errors to fix.

EY survey via Paycom, US, 2022

2.84

salary claims per 1,000 employees in 2025

Up from 2.63 in 2024. Every claim is a pay problem an employee had to escalate to TADM.

MOM Employment Standards Report 2025

S$8,000

new CPF wage ceiling from 1 Jan 2026

CPF, MOM and IRAS rules change every year. Each change needs settings updated in time.

CPF Board

Payment-diversion fraud is rising in Singapore

Business email compromise, where attackers redirect payments to their own account, was the **3rd-largest scam type by losses** in the first half of 2026. Reported losses were **S$57.3M**, up from S$19.5M a year earlier. Payroll diversion uses the same trick on salaries: a stolen login changes the bank account. Most HR tools leave the defence as advice, not as checks inside payroll.

Singapore Police Force mid-year statistics 2026; S$ figures as reported by Scam.SG. Dayforce payroll diversion guidance.

The market gap (checked October 2026)

- ADP, Workday and Deel check pay before payout. Singapore vendors we checked do not document it.
- None of the 8 vendors documents a hold or cooling-off on new bank details.
- None shares fraud intelligence across customers. Banks do (MAS COSMIC, Cifas, Plaid Beacon).
- None documents human-verified tracking of CPF, MOM and IRAS changes.

See the full vendor table →

WHY · MARKET CHECK

What payroll tools offer today, checked October 2026

Eight vendors, five capabilities, from public product and help pages. ● documented · ◖ partly · — no public evidence found (which does not prove a feature is absent).

VENDOR	CHECK BEFORE PAYOUT	BANK-CHANGE PROTECTION	ATTENDANCE FRAUD CONTROLS	STATUTORY RULE UPDATES	FRAUD INTEL SHARED ACROSS CUSTOMERS
ADP	● ADP Assist flags variances	◖ code to change; advice page	● face-recognition kiosk	◖ monitors rules; CPF not named	—
Rippling	◖ time data only	◖ alert workflow, no hold	● GPS radius	◖ tax updates; CPF not named	—
Deel	● Payroll Detective	—	◖ location check	◖ compliance alerts	—
Workday	● Pay Anomalies (ML)	◖ re-authentication	● geofence	● auto tax updates; SG not shown	—
Dayforce	◖ announced 2023, limited	◖ email alert, form lock	● fingerprint clocks	◖ SG payroll since 2024	—
Talenox (SG)	—	—	—	◖ “updated with regulations”	—
Info-Tech (SG)	—	—	● face, geofence, fake-GPS block	◖ “regularly updated”	—
QuickHR (SG)	—	—	◖ face; any location	—	—
PayShield	● Trust Check on every record	● hold, second channel, cooling-off	● geofence, liveness, device	● AI drafts, person verifies, what-if	● Fraud Network, opt-in

No one holds a hijacked salary

The strongest controls found are alerts and re-authentication. No vendor documents a hold or cooling-off on new bank details.

No one shares fraud intel

Banks already do it: MAS COSMIC (Singapore, 2024), Cifas (UK), Plaid Beacon (US). No payroll vendor applies the idea to salary accounts.

No one shows CPF tracking

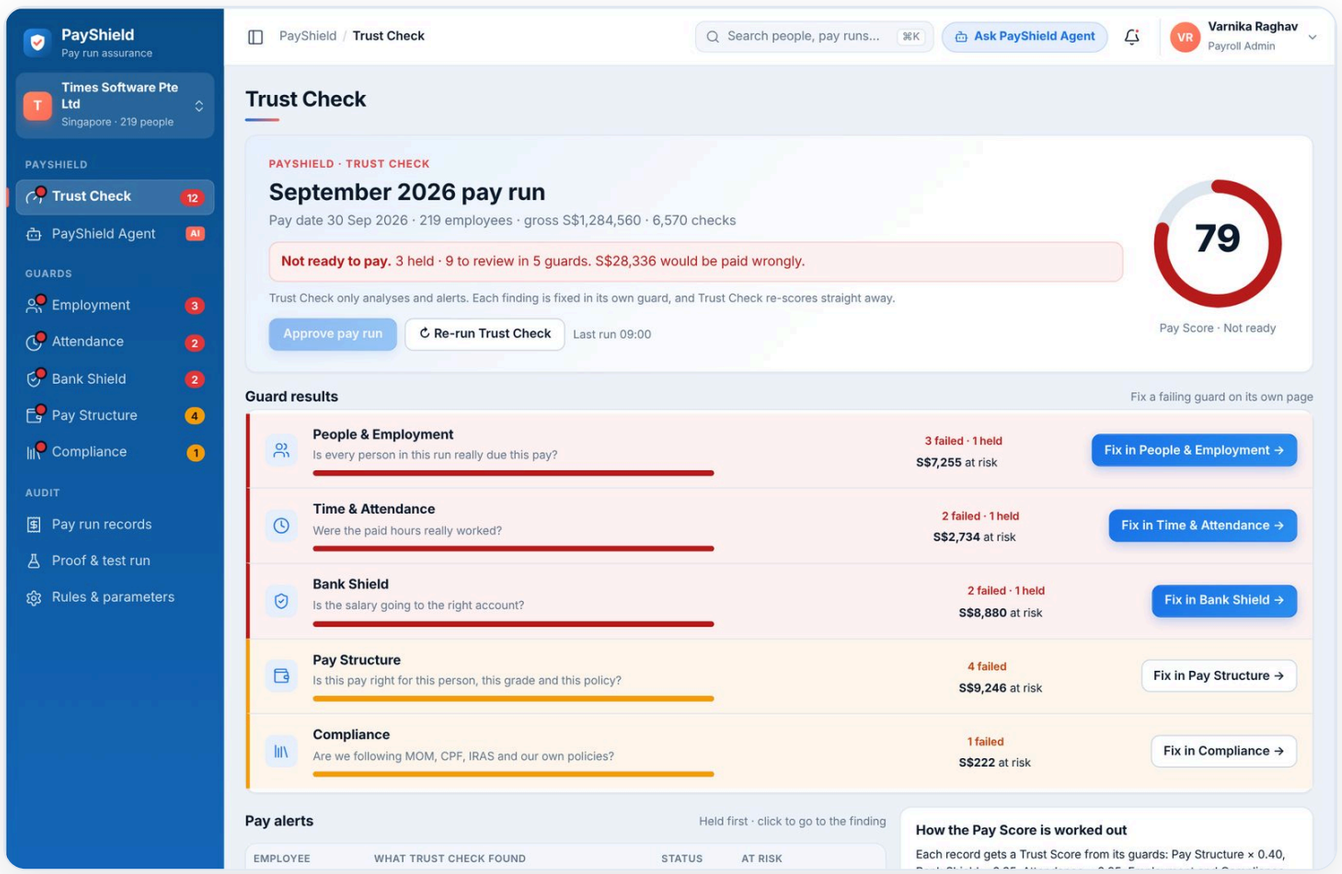
Singapore vendors say rules are “updated”. None documents automatic, human-verified tracking of CPF, MOM and IRAS changes.

Sources: adp.com and ADP press (Sep 2025, Apr 2026); rippling.com; deel.com and CPA Practice Advisor (Aug 2025); doc.workday.com; dayforce.com; talenox.com; info-tech.com.sg; quickhr.co; fenergo.com (MAS COSMIC); fca.org.uk (Cifas); plaid.com/products/beacon. Some evidence comes from customer guides or press, not vendor documentation. Confirm with each vendor before quoting externally.

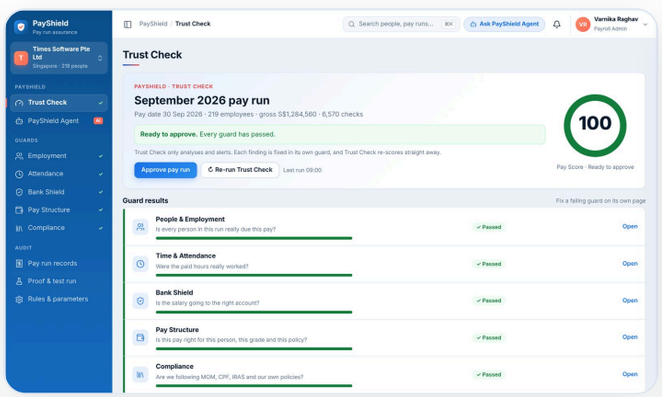
THE HERO PRODUCT

PayShield Trust Check: analyse the run, fix in the guards

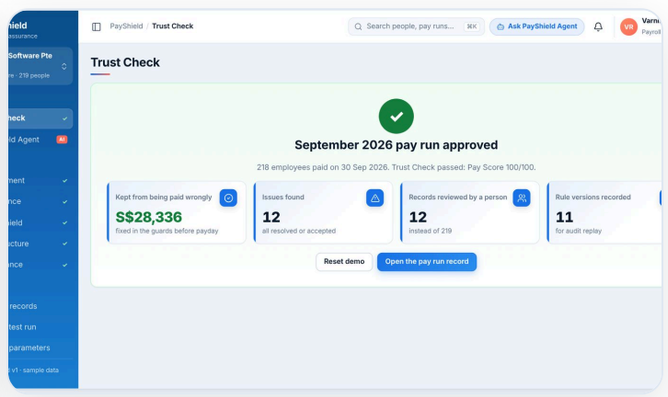
Trust Check sits between “pay run calculated” and “money sent”. It runs all five guards, scores every record and the whole run, and raises alerts. It never fixes anything itself: each alert opens the guard where the finding belongs, and Trust Check re-scores the moment it is fixed.



Trust Check: Pay Score 79, three guards failing with held records. Each row opens its guard.



After fixes in the guards: Pay Score 100, Approve unlocked



Approved: S$28,336 kept from being paid wrongly

How a pay run flows

1. **Calculate** the pay run as today
2. **Trust Check** runs the five guards on all 219 records and gives a Pay Score
3. **Alerts** show which guard failed, held first, with the amount at risk
4. **Click an alert** to open that guard at the finding; fix or accept it there with a reason
5. **Trust Check re-scores**; Approve unlocks at 100

How PayShield decides

Risk = **0.40** × Pay Structure + **0.35** × Bank Shield + **0.25** × Attendance, +0.03 per extra strong signal. Employment and Compliance breaches are critical rules: at least Review.

80+ Clear | 50–79 Review | under 50 Held

What the payroll team gets

- One page that says “ready to pay” or exactly why not
- 12 records to look at instead of 219
- A red dot on each guard with work pending, and one place to fix each kind of problem
- Every decision and rule version recorded for audit

A person always decides

PayShield never changes pay by itself. It holds, explains and suggests; an admin or approver acts.

THE FIVE GUARDS

Five guards feed one decision

Each guard is its own page in PayShield, with its own findings, validators and parameters. Together they answer: is this person due this pay, in this account, under today's rules?

1

People & Employment

Is every person in this run really due this pay?

- New joiners and proration
- Leavers and final pay
- Work pass validity
- Contract and employment changes

In the sample run: 3 found

Open the guard →

2

Time & Attendance

Were the paid hours really worked?

- Overtime cap and rate
- Paid hours vs clock-ins
- Buddy punching (geofence, selfie, one phone)
- Missed clock-outs
- OT pre-approval policy

In the sample run: 2 found

Open the guard →

3

Bank Shield

Is the salary going to the right account?

- Bank changes near payday
- New device sign-ins
- Second-channel confirmation
- Shared accounts
- Fraud Network across companies

In the sample run: 2 found

Open the guard →

4

Pay Structure

Is this pay right for this person, this grade and this policy?

- Salary vs own history
- Grade bands
- Increments and approvals
- Bonuses
- AWS (13th month)
- Stock awards

In the sample run: 4 found

Open the guard →

5

Compliance

Are we following MOM, CPF, IRAS and our own policies?

- Statutory rule library (versioned)
- Rule Watch: new rules drafted from official sources
- Company policies
- Audit trail of rule versions per pay run

In the sample run: 1 found

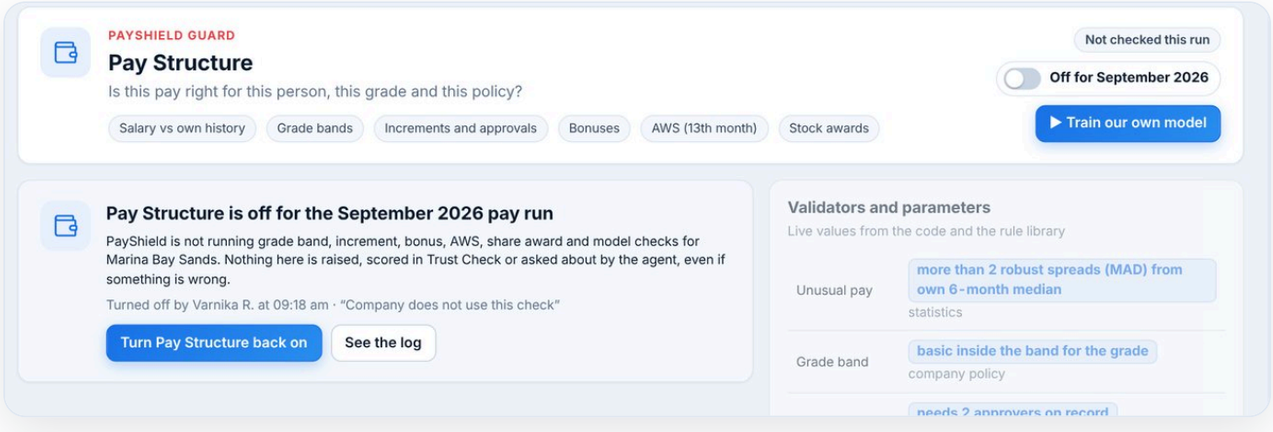
Open the guard →

Built from what Times HRMS already holds

Employee records, timesheets, bank details, pay items and statutory settings. PayShield reads them; it does not need a new data source.

Sold as one product

Customers buy PayShield, not five tools. Guards can be switched on in stages during rollout, starting with Bank Shield and Compliance.



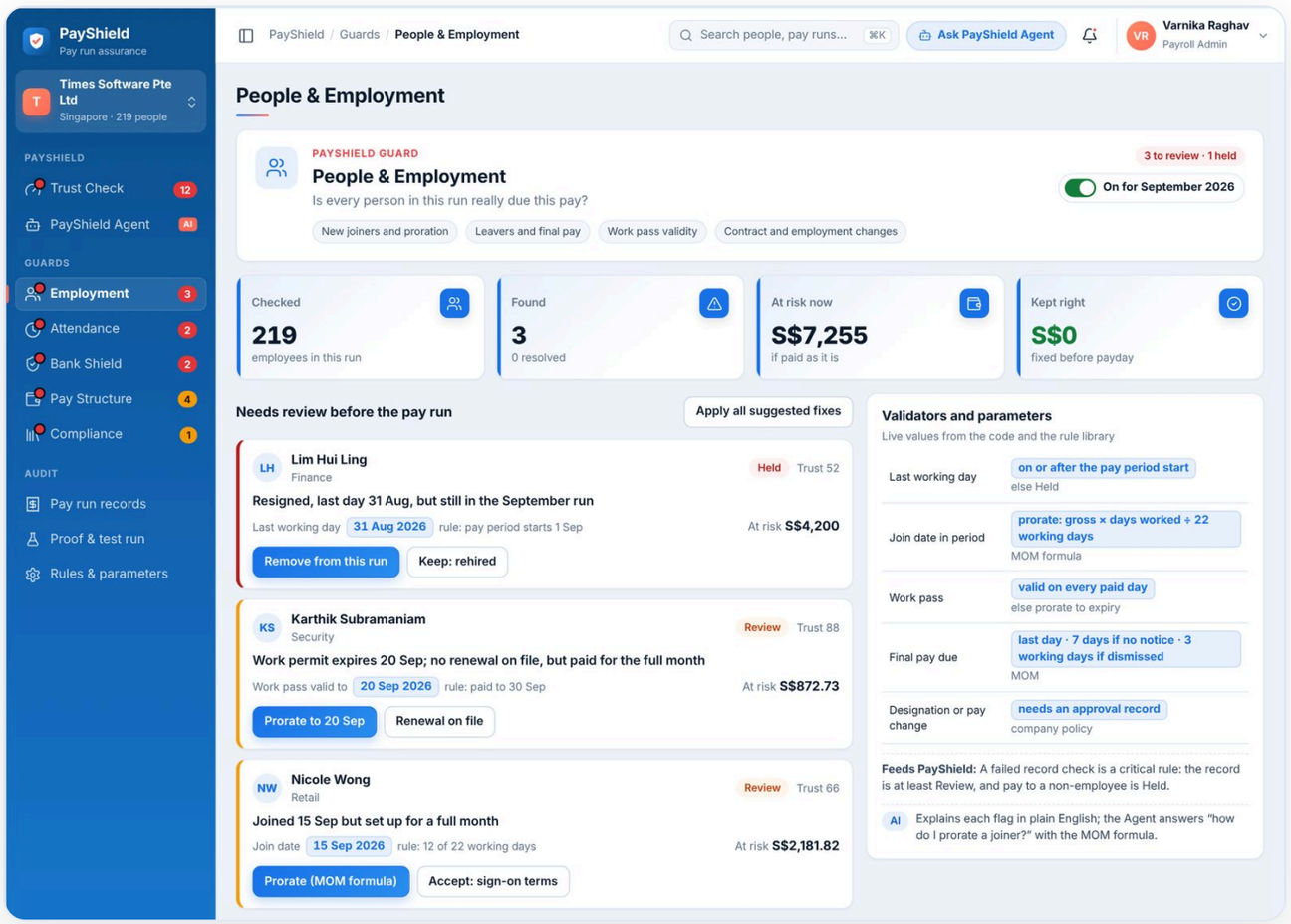
Pay Structure switched off for the September 2026 run: its checks are skipped and the reason is logged

Each guard can be switched off for a pay run

- A toggle on every guard page: **this pay run only** or **until turned back on**
- A reason is required; Bank Shield and Compliance also need a second person to approve
- While off, its checks do not run: nothing is raised, scored in Trust Check or asked about by the agent, even if something is wrong
- Trust Check shows the guard as "Off · not checked"; the agent reports it as skipped
- Every switch is logged in Pay run records with who, when, reason and scope

GUARD 1 OF 5

People & Employment: is every person in this run really due this pay?



People & Employment page in PayShield: findings, validators and parameters

Found in the sample September run · 3 records · S$7,255 at risk

Lim Hui Ling · Resigned, last day 31 Aug, but still in the September run
Last working day: **31 Aug 2026** · rule: pay period starts 1 Sep → *Remove from this run*
Held · 52 — **S$4,200**

Karthik Subramaniam · Work permit expires 20 Sep; no renewal on file, but paid for the full month
Work pass valid to: **20 Sep 2026** · rule: paid to 30 Sep → *Prorate to 20 Sep*
Review · 88 — **S$873**

Nicole Wong · Joined 15 Sep but set up for a full month
Join date: **15 Sep 2026** · rule: 12 of 22 working days → *Prorate (MOM formula)*
Review · 66 — **S$2,182**

REAL-WORLD PROBLEM

Leavers still paid, joiners paid a full month, work passes expiring mid-month.

These are usually found at month-end reconciliation, after the money has gone, and recovering it is awkward for everyone.

WITH PAYSHIELD

No payments to leavers. Joiners and expiring work passes are prorated with the MOM formula before payday, not fixed after it.

Validators and parameters

Last working day	**on or after the pay period start** · else Held
Join date in period	**prorate: gross × days worked ÷ 22 working days** · MOM formula
Work pass	**valid on every paid day** · else prorate to expiry
Final pay due	**last day · 7 days if no notice · 3 working days if dismissed** · MOM
Designation or pay change	**needs an approval record** · company policy

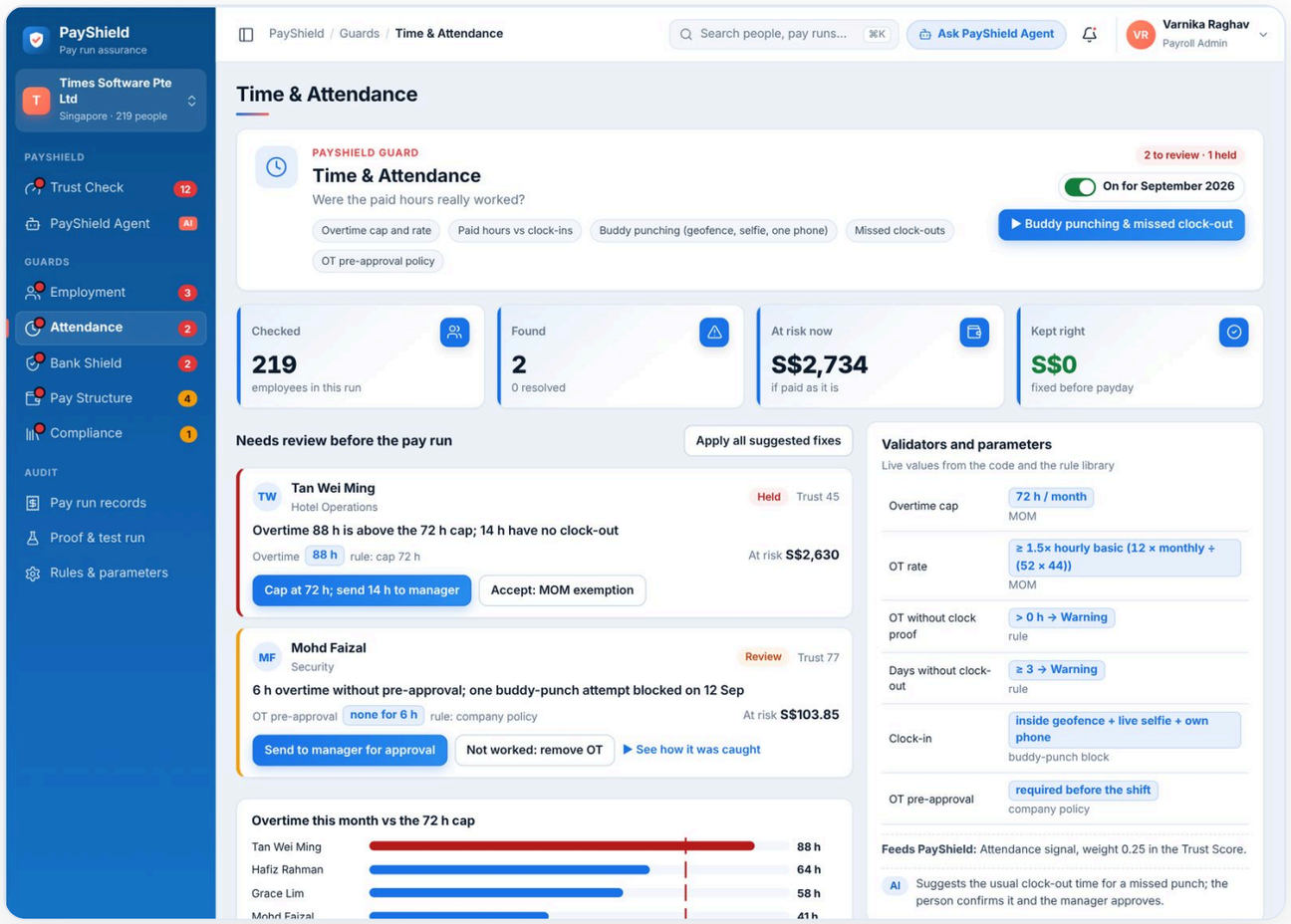
Feeds PayShield: A failed record check is a critical rule: the record is at least Review, and pay to a non-employee is Held.

AI: Explains each flag in plain English; the Agent answers "how do I prorate a joiner?" with the MOM formula.

Try it in the demo: Open from a Trust Check alert

GUARD 2 OF 5

Time & Attendance: were the paid hours really worked?



Time & Attendance page in PayShield: findings, validators and parameters

Found in the sample September run · 2 records · S$2,734 at risk

Tan Wei Ming · Overtime 88 h is above the 72 h cap; 14 h have no clock-out
Overtime: **88 h** · rule: cap 72 h → *Cap at 72 h; send 14 h to manager*
Held · 45 · **S$2,630**

Mohd Faizal · 6 h overtime without pre-approval; one buddy-punch attempt blocked on 12 Sep
OT pre-approval: **none for 6 h** · rule: company policy → *Send to manager for approval*
Review · 77 · **S$104**

REAL-WORLD PROBLEM

Overtime above the legal cap, hours with no clock-in, and buddy punching.

Overtime is paid on what the timesheet says. Without proof behind the hours, overpayment and MOM breaches go unseen.

WITH PAYSHIELD

Only proven hours are paid. Overtime above the cap goes to the manager, not onto the payslip, and one phone can clock in only one person.

Validators and parameters

Overtime cap	**72 h / month** · MOM
OT rate	**≥ 1.5× hourly basic (12 × monthly ÷ (52 × 44))** · MOM
OT without clock proof	**> 0 h → Warning** · rule
Days without clock-out	**≥ 3 → Warning** · rule
Clock-in	**inside geofence + live selfie + own phone** · buddy-punch block
OT pre-approval	**required before the shift** · company policy

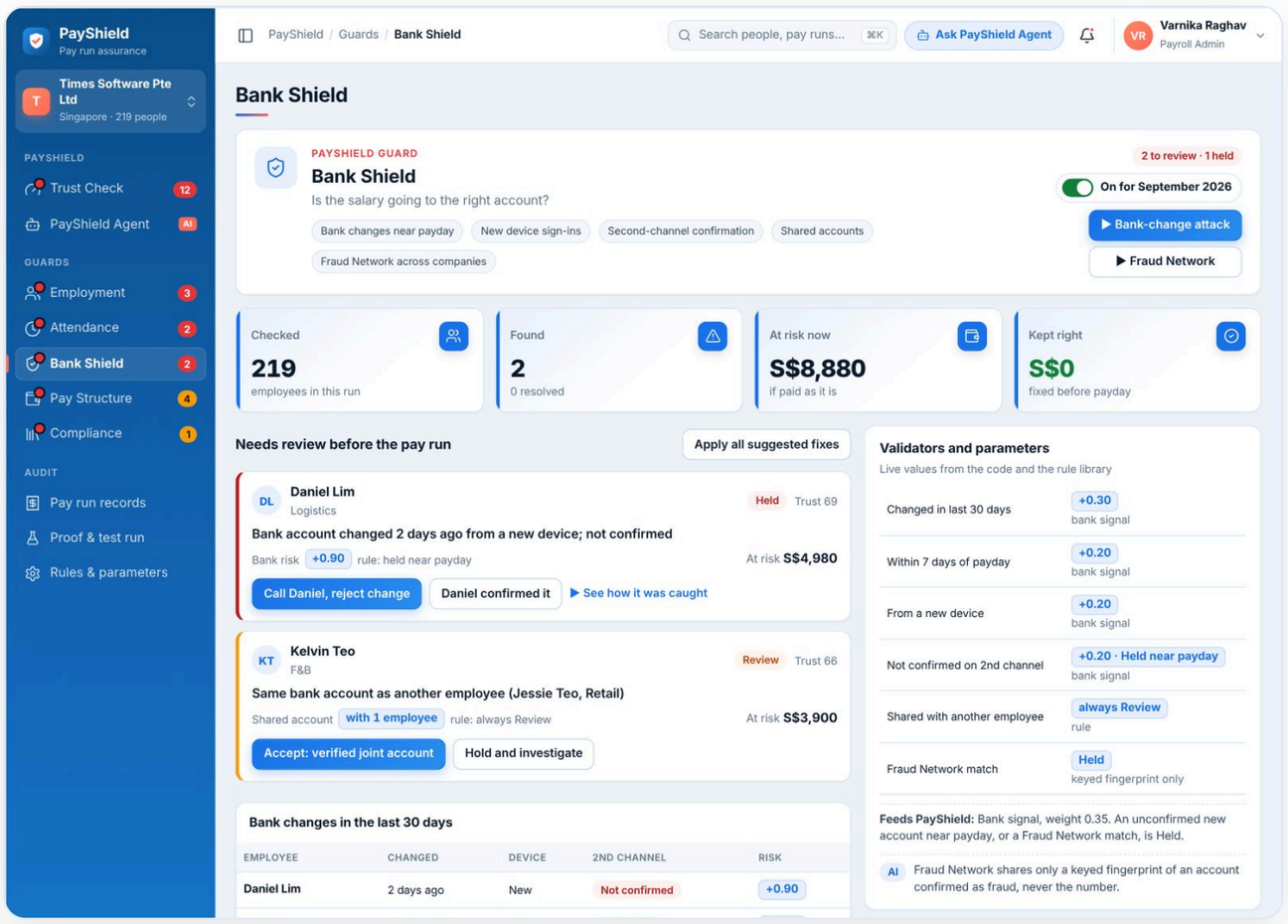
Feeds PayShield: Attendance signal, weight 0.25 in the Trust Score.

AI: Suggests the usual clock-out time for a missed punch; the person confirms it and the manager approves.

Try it in the demo: Buddy punching & missed clock-out

GUARD 3 OF 5

Bank Shield: is the salary going to the right account?



Bank Shield page in PayShield: findings, validators and parameters

Found in the sample September run · 2 records · S$8,880 at risk

Daniel Lim · Bank account changed 2 days ago from a new device; not confirmed
Bank risk: **+0.90** · rule: held near payday → *Call Daniel, reject change*
Held · 69
S$4,980

Kelvin Teo · Same bank account as another employee (Jessie Teo, Retail)
Shared account: **with 1 employee** · rule: always Review → *Accept: verified joint account*
Review · 66
S$3,900

REAL-WORLD PROBLEM

A stolen password is enough to change a bank account a day before payday.

The salary goes to the attacker, the employee is paid again off-cycle, and the money is rarely recovered.

WITH PAYSHIELD

A changed account is paid only after the employee confirms on a second channel. Accounts confirmed as fraud are blocked at every company on the Fraud Network.

Validators and parameters

Changed in last 30 days	**+0.30** · bank signal
Within 7 days of payday	**+0.20** · bank signal
From a new device	**+0.20** · bank signal
Not confirmed on 2nd channel	**+0.20 · Held near payday** · bank signal
Shared with another employee	**always Review** · rule
Fraud Network match	**Held** · keyed fingerprint only

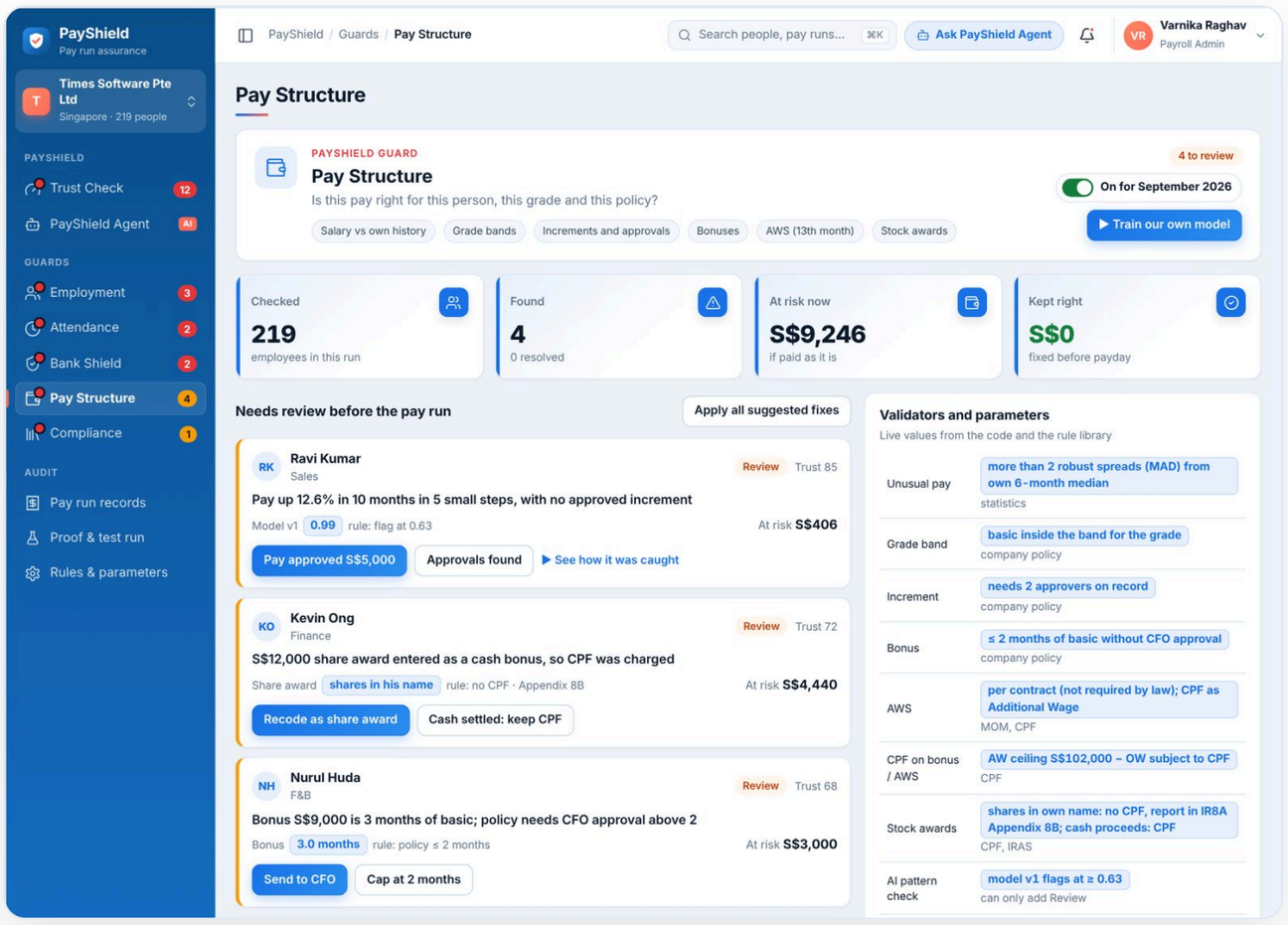
Feeds PayShield: Bank signal, weight 0.35. An unconfirmed new account near payday, or a Fraud Network match, is Held.

AI: Fraud Network shares only a keyed fingerprint of an account confirmed as fraud, never the number.

Try it in the demo: Bank-change attack · Fraud Network

GUARD 4 OF 5

Pay Structure: is this pay right for this person, this grade and this policy?



Pay Structure page in PayShield: findings, validators and parameters

Found in the sample September run · 4 records · S$9,246 at risk

Ravi Kumar · Pay up 12.6% in 10 months in 5 small steps, with no approved increment
Model v1: **0.99** · rule: flag at 0.63 → *Pay approved S$5,000*
Review · 85 — **S$406**

Kevin Ong · S$12,000 share award entered as a cash bonus, so CPF was charged
Share award: **shares in his name** · rule: no CPF · Appendix 8B → *Recode as share award*
Review · 72 — **S$4,440**

Nurul Huda · Bonus S$9,000 is 3 months of basic; policy needs CFO approval above 2
Bonus: **3.0 months** · rule: policy ≤ 2 months → *Send to CFO*
Review · 68 — **S$3,000**

Marcus Lee · Basic S$7,900 is above the grade R3 band (S$4,200–S$6,500)
Grade band R3: **S$7,900** · rule: max S$6,500 → *Attach promotion letter*
Review · 76 — **S$1,400**

REAL-WORLD PROBLEM

Unapproved raises in small steps, bonuses outside policy, share awards with CPF charged, salaries outside the grade.

Each one is small or looks normal on its own, so no single rule catches it; the cost builds up over months.

WITH PAYSHIELD

Raises, bonuses, AWS and share awards follow policy and the right CPF treatment. Slow salary creep is flagged by our own model.

Validators and parameters

Unusual pay	**more than 2 robust spreads (MAD) from own 6-month median** · statistics
Grade band	**basic inside the band for the grade** · company policy
Increment	**needs 2 approvers on record** · company policy
Bonus	**≤ 2 months of basic without CFO approval** · company policy
AWS	**per contract (not required by law); CPF as Additional Wage** · MOM, CPF
CPF on bonus / AWS	**AW ceiling S$102,000 – OW subject to CPF** · CPF
Stock awards	**shares in own name: no CPF, report in IR8A Appendix 8B; cash proceeds: CPF** · CPF, IRAS
AI pattern check	**model v1 flags at ≥ 0.63** · can only add Review

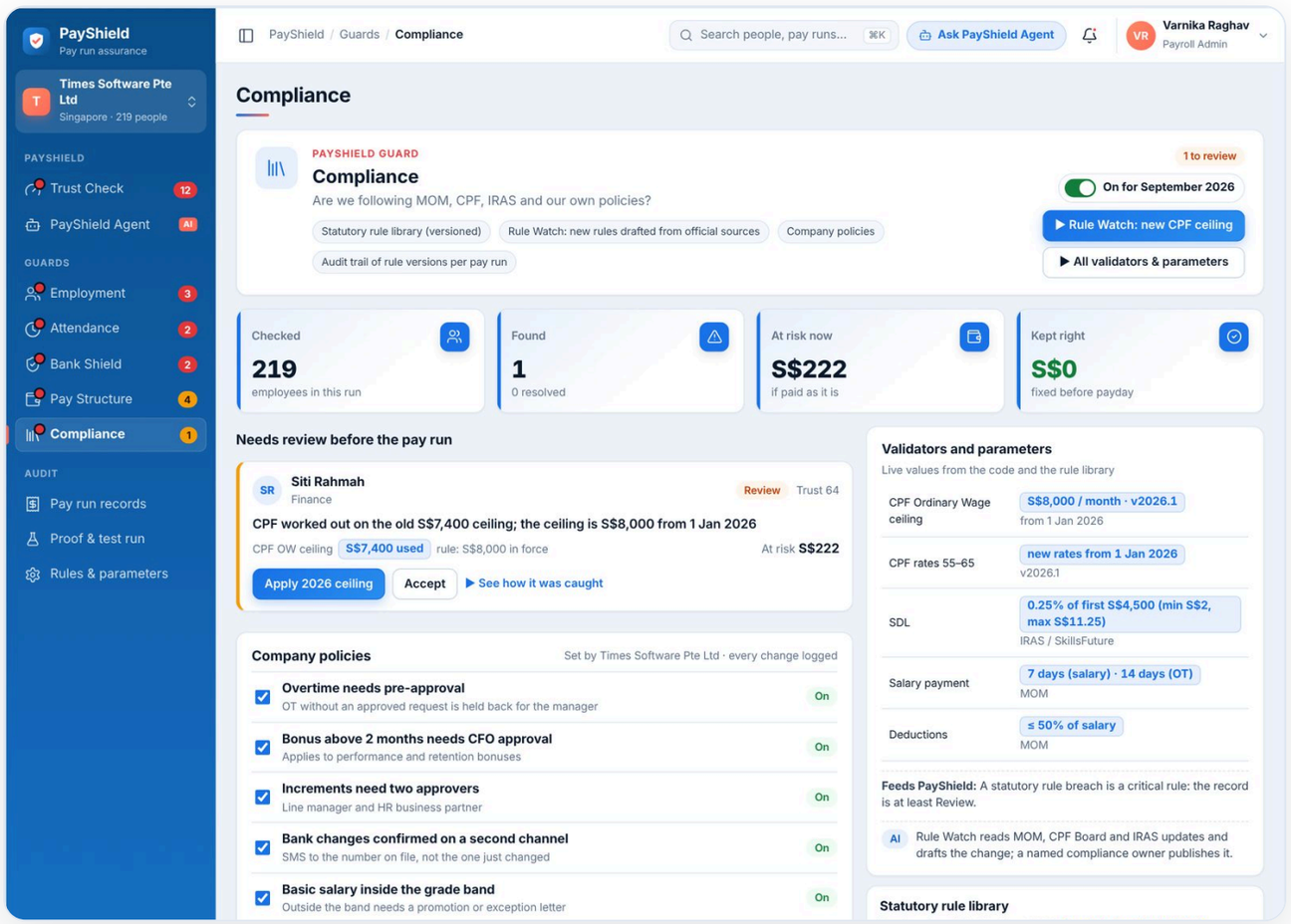
Feeds PayShield: Pay signal, weight 0.40. Our own model can add Review for slow patterns.

AI: Our own model finds salary creep and raises far above peers that no single rule catches.

Try it in the demo: Train our own model

GUARD 5 OF 5

Compliance: are we following MOM, CPF, IRAS and our own policies?



Compliance page in PayShield: findings, validators and parameters

Found in the sample September run · 1 record · S$222 at risk

Siti Rahmah · CPF worked out on the old S$7,400 ceiling; the ceiling is S$8,000 from 1 Jan 2026
CPF OW ceiling: **S$7,400 used** · rule: S$8,000 in force → *Apply 2026 ceiling*

Review · 64
S$222

REAL-WORLD PROBLEM

CPF ceilings, rates and MOM rules change, and company policies are kept in people's heads.

A missed setting shows up as short CPF months later, with back-payment and late-payment interest.

WITH PAYSHIELD

Every run uses today's verified statutory rules and the company's own policies, and records which versions it used for audit.

Validators and parameters

CPF Ordinary Wage ceiling	**S$8,000 / month · v2026.1** · from 1 Jan 2026
CPF rates 55–65	**new rates from 1 Jan 2026** · v2026.1
SDL	**0.25% of first S$4,500 (min S$2, max S$11.25)** · IRAS / SkillsFuture
Salary payment	**7 days (salary) · 14 days (OT)** · MOM
Deductions	**≤ 50% of salary** · MOM

Feeds PayShield: A statutory rule breach is a critical rule: the record is at least Review.

AI: Rule Watch reads MOM, CPF Board and IRAS updates and drafts the change; a named compliance owner publishes it.

Try it in the demo: Rule Watch · All validators

ONLY IN PAYSHIELD

Two features no payroll tool we reviewed offers

Both can be demonstrated today with sample data. Neither needs real customer data for the ideathon.

★ FRAUD NETWORK

Fraud Network: every company protects the others

Company A
Confirms a bank account was used in fraud

→

Fraud Network
Keeps only a keyed fingerprint

→

Company B
Same account is blocked

Privacy
Only a keyed one-way fingerprint (HMAC) of the account number is shared, never the number itself. No company names or employee data are shared, and customers agree to this in their contract.

Rollout: opt-in, starting inside multi-company groups; legal and PDPA review before any sharing between unrelated customers.

★ WHAT-IF FOR RULE WATCH

See a rule's cost before it starts

Rule Watch reruns the company's last pay run in a sandbox with the new rule. Nothing is saved; it only shows the difference.

Example: CPF wage ceiling S$7,400 → S$8,000 from 1 Jan 2026
Run on Acme's November 2025 payroll, six weeks before the change (sample data)

46	+S$4,210	+S$50,520
employees affected	employer CPF a month	over 12 months



Extra employer CPF per month by pay group · totals 46 people and S$4,210

Why it matters: Rule Watch stops being only an alert and becomes a planning tool. Finance can budget the change, and HR can tell employees before their take-home pay moves.

Still human-approved
The rule behind every what-if has been verified by a compliance owner, and the company admin confirms any setting change.

WHERE THE AI IS

Five jobs for AI, and one it deliberately does not do

AI does the work that needs judgement or language. The score that holds a salary stays plain arithmetic, so every hold can be explained to an employee, an auditor or MOM.

1

Learns from every decision

Fewer false alarms every month

- An accepted reason (“promotion”) is remembered for 3 pay runs, then expires
- In the test run, repeat alarms fell from 6.1 to 0
- Spots patterns: “pay spike on a promotion accepted 9 of 10 times → ask for a reason code at entry”

Guardrail: it only suggests. An admin approves every change, and it never switches off rule or bank checks.

2

Reads rule changes

Rule Watch

- Reads MOM, CPF Board and IRAS updates on a schedule
- Drafts the change: value, effective date, source
- What-if shows each company the cost before the rule starts

Guardrail: a named compliance owner verifies value, date and source before anything is published.

3

Explains every flag

Plain English for the admin

- One sentence per flagged record: “Bank changed 2 days before payday from a new device”
- Built from a masked summary: no names, NRIC or bank numbers
- Every number checked against the record

Guardrail: if the AI is off or wrong, a fixed template sentence is used. Payroll never depends on AI.

Deliberately not AI: the score

The Trust Score is rules plus statistics on each person’s own history. It is predictable, testable (50 unit tests) and explainable line by line. A black-box model should not be the thing that holds someone’s salary.

4 · Our own trained model

PayShield Risk Model v1 re-checks what the rules cleared and finds slow patterns such as salary creep and raises far above peers. It runs inside our system and can only add a Review flag. **See the results →**

5 · PayShield Agent

One prompt runs every guard, our own model, compliance and history, and returns a Pay Score with a fix for every finding. People confirm and approve; every run is recorded. It also answers payroll questions from 41 verified answers. **See how →**

WHERE THE AI IS · OUR OWN MODEL

Our own model finds what rules miss: 120 of 120 salary-increase cases

PayShield Risk Model v1 is a small model we train ourselves on de-identified payroll features. It looks again at records the rules cleared, and it can only add a Review flag.

RULES ALONE
500 / 620
hidden problems caught in 20 companies the model never saw

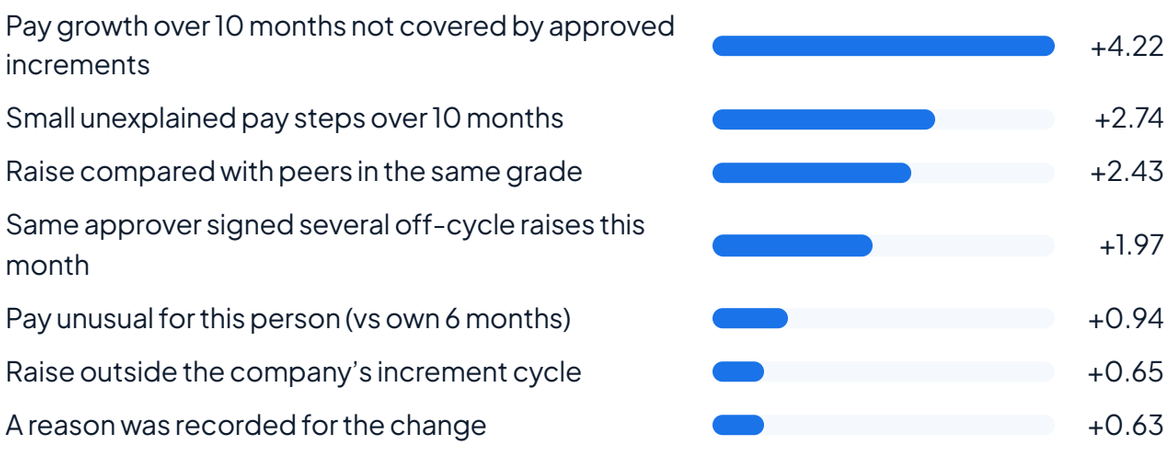
RULES + OUR MODEL
620 / 620
salary creep and raises far above peers: 0 → 120

HARDER TEST
103 / 120
smaller creep steps, smaller raises, two approvers. We report this too.

FALSE ALARMS
2.4%
of clean records, from 1.6% with rules alone

What it learned, from 9,344 records

Top weights of 15 inputs. Each flag shows its top 3 reasons, e.g. “pay growth over 10 months not covered by approved increments”.



Security by design

- **Runs inside our system.** Training and scoring need no outside AI service.
- **15 allow-listed inputs.** No ids, names, NRIC, bank numbers, gender, age, race or nationality. The build fails if one is added.
- **Can only add Review.** It never clears a rule flag, holds pay or changes pay.
- **Hard to poison.** Retrained only from two-person decisions; each version needs two approvers.
- **Drift alarm.** If its flag rate leaves 0.5–3%, it switches to rules-only.

Two models of our own, two jobs

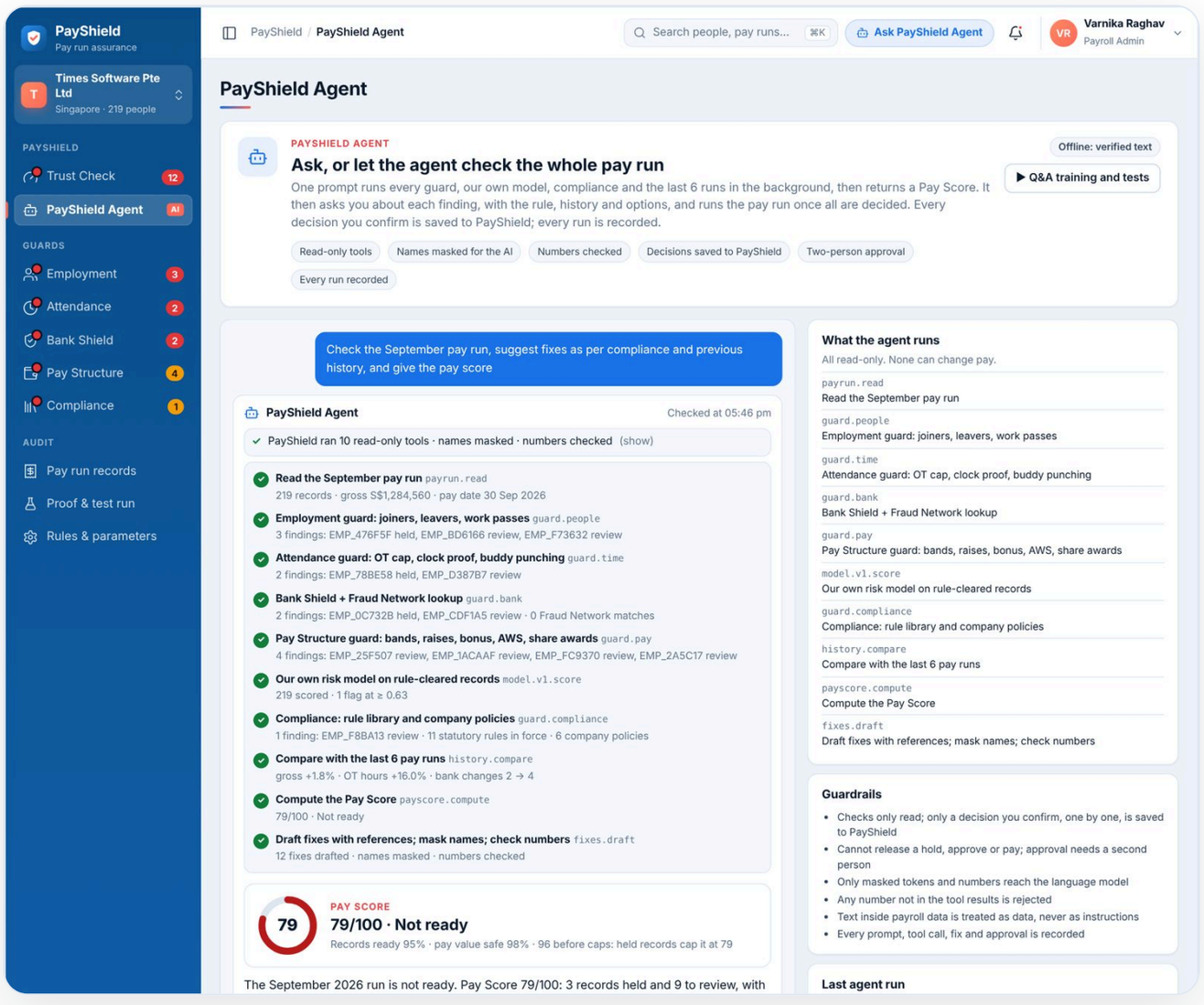
This small risk model finds the patterns and can be explained input by input. PayShield Agent, our payroll language model, answers questions and words the results, from masked data only. See the agent →

Synthetic data: we planted these patterns, so this shows the model can learn them, not how it performs on real payroll. Next: retrain on pilot customers’ audited decisions. Reproduce with `npm test` or the demo’s “Our own model” tab.

WHERE THE AI IS · PAYSHIELD AGENT

One prompt runs the whole of PayShield

"Check the September pay run, suggest fixes as per compliance and previous history, and give the pay score." The agent runs all of PayShield in the background and answers with a Pay Score and a fix for every finding.



PayShield Agent in the prototype: 10 read-only tools, then Pay Score 79/100 and 12 fixes with their rule and history

What runs in the background

`payrun.read`
Read the pay run

`guard.time`
Attendance guard

`guard.pay`
Pay Structure guard

`guard.compliance`
Rule library + company policies

`payscore.compute`
Pay Score

`guard.people`
Employment guard

`guard.bank`
Bank Shield + Fraud Network

`model.v1.score`
Our own risk model

`history.compare`
Last 6 pay runs

`fixes.draft`
Fixes with rule and history

All read-only. None of them can change pay.

What comes back

- **Pay Score** for the whole run: half records ready, half pay value safe. Held records cap it at 79; open reviews at 89.
- **Every finding** with its rule (MOM, CPF, IRAS or company policy) and the employee's history
- **A suggested fix** and the amount at risk

Then it asks, you decide, it runs

Agent checks → Asks about each finding → You choose A, B or C → Saved to the guard → Approver signs off → Agent runs the pay run → Recorded

Each choice is saved to its guard as you go, so PayShield is up to date before the pay run starts.

WHERE THE AI IS · PAYSHIELD AGENT

It asks about each finding, then runs the pay run

Held findings first. For each one the agent shows what it found, the rule, the employee's history and the money at risk, recommends an option and explains why. You choose A, B or C, and the choice is saved to its guard in PayShield. When all are decided, an approver signs off and the agent runs the pay run and records it.

PayShield Agent — Finding 1 of 12 · Bank Shield

Held **Daniel Lim** Logistics · Trust 69

Bank account changed 2 days ago from a new device; not confirmed.

CHECKED Bank risk: **+0.90** · rule: held near payday
RULE Company policy: bank changes confirmed on a second channel
HISTORY Same OCBC account for 4 years; never changed before
AT RISK **S$4,980** if paid as it is

How do you want to fix this? I recommend **A**, because the rule is "Company policy: bank changes confirmed on a second channel", and the history shows: same OCBC account for 4 years; never changed before.

A **Call Daniel, reject change** — Recommended
Daniel: "Not me." Change rejected; pay goes to his usual OCBC ···· 2209. Fingerprint shared with the Fraud Network.

B **Daniel confirmed it**
Confirmed by Daniel on his number on file; new account accepted. Choose this only if you have the evidence.

C **Leave out of this run**
Daniel is not paid in this run; Bank Shield records the hold, and pay goes off-cycle once the evidence is in.

Your choice is saved to Bank Shield in PayShield. You can also type A, B or C, or ask me a question first.

Finding 1 of 12: Daniel Lim, Bank Shield, held. Options with what each does to this run.

PayShield Agent — All findings decided

100 PAY SCORE
100/100 · Ready to approve
All 12 decisions saved to PayShield · Trust Check 100/100

11 fixed as recommended, 0 accepted with a reason, 1 left out of this run. 217 employees paid · 1 left out of this run · S$24,436 kept from being paid wrongly.

EMPLOYEE	GUARD	YOUR DECISION	SAVED TO
Daniel Lim	Bank	**A** · Call Daniel, reject change	Bank Shield
Lim Hui Ling	People	**A** · Remove from this run	People & Employment
Tan Wei Ming	Time	**A** · Cap at 72 h; send 14 h to manager	Time & Attendance
Kevin Ong	Pay	**A** · Recode as share award	Pay Structure
Kelvin Teo	Bank	**C** · Leave out of this run	Bank Shield
Nurul Huda	Pay	**A** · Send to CFO	Pay Structure
Nicole Wong	People	**A** · Prorate (MOM formula)	People & Employment
Marcus Lee	Pay	**A** · Attach promotion letter	Pay Structure
Karthik Subramaniam	People	**A** · Prorate to 20 Sep	People & Employment
Ravi Kumar	Pay	**A** · Pay approved S$5,000	Pay Structure
Siti Rahmah	Compliance	**A** · Apply 2026 ceiling	Compliance
Mohd Faizal	Time	**A** · Send to manager for approval	Time & Attendance

✓ PayShield is up to date: 12 decisions saved to their guards. Trust Check and the guard pages already show them.

Run the September 2026 pay run

All 12 decided: Pay Score 100, each decision saved to its guard

Every decision is saved to PayShield

- Each A, B or C you confirm is **saved to its guard straight away**; Trust Check re-scores after every answer (79 → 100)
- The guard page shows "Decided with PayShield Agent" with who and when, plus Undo
- No temporary fixes: Trust Check, the guards and the pay run always agree
- Only a decision the user confirms, one at a time, can change data; a prompt alone cannot

PayShield Agent — Pay run completed at 11:19 pm

✓ **Lock inputs and rule versions for September 2026** `payrun.lock`
✓ **Use the 12 decisions already saved to PayShield** `decisions.apply`
✓ **Calculate pay for 217 employees** `pay.calculate`
✓ **Prepare CPF and SDL submission** `cpf.file`
✓ **Prepare the bank (GIRO) file for 217 employees** `bank.file`
✓ **Generate 217 payslips** `payslips`
✓ **Write the pay run record** `audit.write`

✓ **September 2026 pay run completed.** 217 employees paid · 1 left out of this run · S$24,436 kept from being paid wrongly. Approved by Grace Lim. Logged as **PR-2026-09** with every question, decision and approval. Open the audit record →

PayShield already has all 12 decisions, so nothing is left to fix for September.
Open Trust Check →

}

Final run: lock inputs, use the saved decisions, calculate, CPF and GIRO files, payslips, audit record

Audit record PR-2026-09 keeps the prompt, every tool call, each question with the option chosen, who chose it and when, the approver and the rule versions. It also answers payroll questions: 41 verified answers, 82% right first time on unseen wording.

PROOF · TEST RUN

We tested it: 25 of 25 hidden problems caught

A made-up company of 500 employees with 12 months of pay, normal noise (increments, OT swings, promotions, bank changes) and 25 hidden problems. The real `scoring.js` checks it. Repeated on 50 random companies.

CAUGHT

25 / 25

in every one of the 50 runs. First design: 18 on average (17–19).

FALSE ALARMS

1.4%

of clean records (worst run 2.1%). First design: 3.6%. Target under 5%.

REPEAT ALARMS

0

the month after admins accept a reason. Without learning: 6.1 per run.

REAL DATA USED

None

Seeded and repeatable: anyone can rerun it with `npm test` or in the demo.

HIDDEN PROBLEM · ALL 50 COMPANIES	FIRST DESIGN	CURRENT
Bank account changed by a hacker 1–2 days before the run	150 / 150	**150 / 150**
Bank account changed by a hacker 2–3 weeks earlier	100 / 100	**100 / 100**
Hacker also controls the employee's email and confirms the alert	5 / 50	**50 / 50**
Ghost employee paid into another employee's account	2 / 100	**100 / 100**
Pay 35–90% above normal with no reason	250 / 250	**250 / 250**
Wrong CPF contribution	200 / 200	**200 / 200**
Ex-employee still in the pay run	150 / 150	**150 / 150**
Overtime paid with no clock-in proof	20 / 150	**150 / 150**
Full pay for days with no clock-out	23 / 100	**100 / 100**

What the test taught us: 4 fixes

1. **Shared account → always Review.** The ghost-employee pattern scored Clear before.
2. **OT paid without clock proof → Warning rule.**
3. **Pay for 3+ days with no clock-out → Warning rule.**
4. **Second-channel confirmation** for a new account from a new device, so a hijacked email cannot approve it. Hourly pay that matches verified hours is no longer flagged, which cut false alarms.

!

Read this honestly

We wrote both the problems and the checks, so 25 of 25 shows coverage of the patterns we know. Next: the same test on anonymised past pay runs from 2 pilot customers.

TRUST SCORE

How the score is calculated, and when a record is reviewed

The score is plain arithmetic from three signals. It is the same logic running in the prototype's `scoring.js`, covered by 16 unit tests.

1 · THREE SIGNALS, EACH 0 TO 1

Pay variance

Highest of: rule severity (Critical 0.9, Warning 0.6, Info 0.3) and distance from the person's own 6-month median. A bonus, increment or proration cancels the distance part.

Bank-change risk

Changed in 30 days +0.3 · within 7 days of payday +0.2 · new device +0.2 · not confirmed +0.2 · shared account +0.5 · Fraud Network match = 1.0

Attendance integrity

Missing punches, OT without clock-in proof, blocked buddy punches, geofence exceptions.

2 · COMBINE

risk = 0.40 × pay + 0.35 × bank + 0.25 × attendance
+ 0.03 for each extra signal ≥ 0.50

Trust Score = 100 × (1 − risk)

Worked example: Daniel (demo)

Pay 0.40 × 0.82	0.328
Bank 0.35 × 0.90	0.315
Attendance 0.25 × 0.40	0.100
Two strong signals	+0.030
Risk	**0.773**
Trust Score	**23 · Held**

3 · DECIDE

80–100 Clear paid as normal

50–79 Review admin fixes or accepts with a reason

0–49 Held second approver releases it

Hard overrides win over the score

Any Critical rule (wrong CPF, ex-employee still paid, OT below 1.5×) → at least Review. Fraud Network match, or a new bank account not confirmed near payday → Held.

Tuning

Weights and thresholds are settings per company. In the pilot, tune so all planted errors are caught while only 3–5% of records are flagged; keep 70%+ of flags acted on.

SECURITY

The AI never sees personal data

Our own rules engine does every check inside the payroll system. The AI only turns the result into one readable sentence, from a masked summary.

INSIDE TIMES PORTAL · PERSONAL DATA STAYS HERE

Payroll data
Pay, bank details, timesheets, contracts

→

Rules engine
Computes signals and the Trust Score

→

Masking
Token + flag codes + 2–3 numbers only

Admin sees Real name + checked sentence + score. The token is swapped back to the name inside Times Portal.

⇄ masked only

AI service (Azure OpenAI)
Writes one sentence. Prompts are not used to train models and are not shared with OpenAI.

```
{
  "subject": "EMP_1AE53A",
  "flags": ["BANK_CHANGED_RECENT",
            "NEW_DEVICE", "PAY_SPIKE"],
  "values": { "days_before_payday": 2,
              "pay_change_pct": 63 },
  "score": 23, "status": "HELD"
}
```

CONTROL	WHAT WE DO	STANDARD
Data minimisation	Allow-listed payload only. Never names, NRIC/FIN, bank numbers, addresses or salary history.	PDPA · OWASP LLM02
Pseudonymisation	New token per employee per pay run (HMAC-SHA256, key in Azure Key Vault). The token map never leaves our database.	ISO/IEC 27001 controls
AI provider	Azure OpenAI in an approved geography, via private endpoint. Apply for modified abuse monitoring so prompts aren't stored.	Microsoft data privacy terms
Safe output	Every number in the AI sentence must match the payload, otherwise a fixed template is used. Free-text fields never enter prompts.	OWASP LLM01 · LLM05 · LLM09
Works without AI	Scores and holds come from the rules engine. If the AI is off, payroll checks still run.	Resilience by design
Access and encryption	Role-based access (managers see only their team), MFA for admins, TLS 1.2+ in transit, AES-256 at rest.	ISO/IEC 27001 controls
Audit and governance	Every score, flag and decision logged; prompts with personal data never logged. Data protection impact assessment before the pilot.	PDPA

SECURITY · BEYOND THE AI

How PayShield stays secure

Payroll data is a prime target. Besides keeping personal data away from the AI, four layers protect the data, the people who can change it, and PayShield itself.

Data where it lives

Even if a server or backup is stolen

- AES-256 encryption at rest, with keys the company controls (Azure Key Vault)
- Extra field-level encryption for NRIC/FIN, bank account and salary
- TLS 1.2+ for everything in transit
- Screens show only the last 4 digits; seeing more needs a reason and is logged
- Each company's data kept apart by the database itself (row-level security)

Accounts and approvals

Even if a password is stolen

- Phishing-resistant MFA for admins: passkeys or an authenticator app, not SMS
- MFA asked again for a bank change, releasing a hold or approving a pay run
- Two-person rule: nobody approves their own pay or their own change
- Least privilege: managers see only their own team
- Short sessions, rate limits and alerts on unusual sign-ins

Monitoring and audit

Even if someone inside misuses access

- Tamper-proof, write-once log of every score, flag and decision
- Personal data never written to logs or prompts
- Alerts on bulk exports, many bank changes at once, or admin actions at odd hours
- No passwords in code: secrets in Key Vault, services use managed identities

Build and compliance

So it stays secure release after release

- Threat model at design time; OWASP ASVS Level 2 as the build checklist
- Code and dependency scanning on every build; penetration test before go-live
- PDPA: impact assessment before the pilot, data kept only as long as needed
- Breach plan: notify the PDPC within 3 calendar days once a breach is assessed notifiable
- ISO/IEC 27001-aligned controls; SOC 2 report as a later goal

★ FRAUD NETWORK

Fingerprints are keyed, not just hashed

A plain hash is not enough: bank account numbers are short, so an attacker could hash every possible number and reverse the list.

DBS ···· 7731
stays with the company

→

HMAC-SHA256
secret key locked in a hardware security module (HSM)

→

9f3a…c71e
only this is shared

Without the key, a leaked fingerprint list is useless. Later phase: **private set intersection**, so companies find matches without either side revealing its own list.

For the ideathon: sample data only

The demo and this proposal use made-up names, amounts and accounts. No real employee or customer data has been used.

BUSINESS CASE

A feature sales can demo, price and defend

PayShield gives sales a clear story: *"We catch payroll errors and fraud before salary is paid, and we keep you in line with Singapore rules."*

WIN NEW DEALS

A demo that sells itself

- Show a fraud being stopped live in 5 minutes
- Singapore-specific checks global tools don't offer
- Answers the CFO's question: "How do we know payroll is right?"

GROW REVENUE

A paid add-on

- Sold as "PayShield" through the existing Plan → Add-ons flow
- Pricing per employee per month, or bundled in a premium plan (to decide)
- What-if helps finance budget each rule change, a reason to buy for CFOs

KEEP CUSTOMERS

Harder to leave

- Fewer payroll complaints, so fewer reasons to switch
- Audit trail and rule history build up inside Times Portal
- Fraud Network: every company protects the others

Best-fit customers

Construction, marine, logistics, F&B, cleaning	Shift work, heavy OT, many sites and foreign workers
SMEs with a small HR team	No in-house compliance expert to track MOM and CPF changes
Multi-entity groups	Many pay runs a month; one bad run is costly

What we will measure in the pilot

METRIC	PROPOSED TARGET
Reruns after lock	–50%
Salaries paid to a fraudulent account	Zero
Demo-to-deal win rate where PayShield is shown	Track vs baseline
Add-on attach rate in new deals	Track from launch

Targets are proposals to confirm with pilot customers and sales leadership.

PRICE BENCHMARK

Singapore payroll list prices per employee per month: Talenox S$5 per extra employee, QuickHR S$3–9, Employment Hero S$6–8, Info-Tech from S$2. Add-ons elsewhere cost about 20–50% of the base price (PeopleHR £1.50–2 on £3–9.50).

Vendor pricing pages, Oct 2026, excl. GST

PROPOSED, FOR SALES TO CONFIRM

S$2.00 per employee per month, **S$1.50** above 200 employees, minimum S$60 a month. 30 customers × 200 employees × S$1.50 = **S$108,000 a year**. A 100-person company pays S$2,400 a year: less than one salary lost to a hijacked bank account.

PLAN AND ASK

Start small, show value in the first month

PHASE 1 · ABOUT 2 SPRINTS

Pay Structure guard + Trust Check

~10 rules on data we already hold, plus the test run repeated on anonymised past pay runs.

PHASE 2 · ABOUT 1 SPRINT

Rule Watch + AI explanations

Versioned rule library with reviewer approval, what-if impact, plain-English reasons on every flag.

PHASE 3 · ABOUT 2 SPRINTS

Bank Shield + Fraud Network

Second-channel confirmation, cooling-off, approvals card. Fraud Network opt-in, inside company groups first.

PHASE 4 · 1–2 SPRINTS

Attendance guard

Geofence, device binding, missed-punch nudges, liveness via an SDK.

What we need from Heads of Departments

- Approval to build Phase 1 and 2
- A small squad: backend, frontend and QA
- A named compliance owner to verify rule updates
- Two pilot customers with shift work and OT, and anonymised past pay runs to re-test on
- Legal and PDPA review of Fraud Network sharing

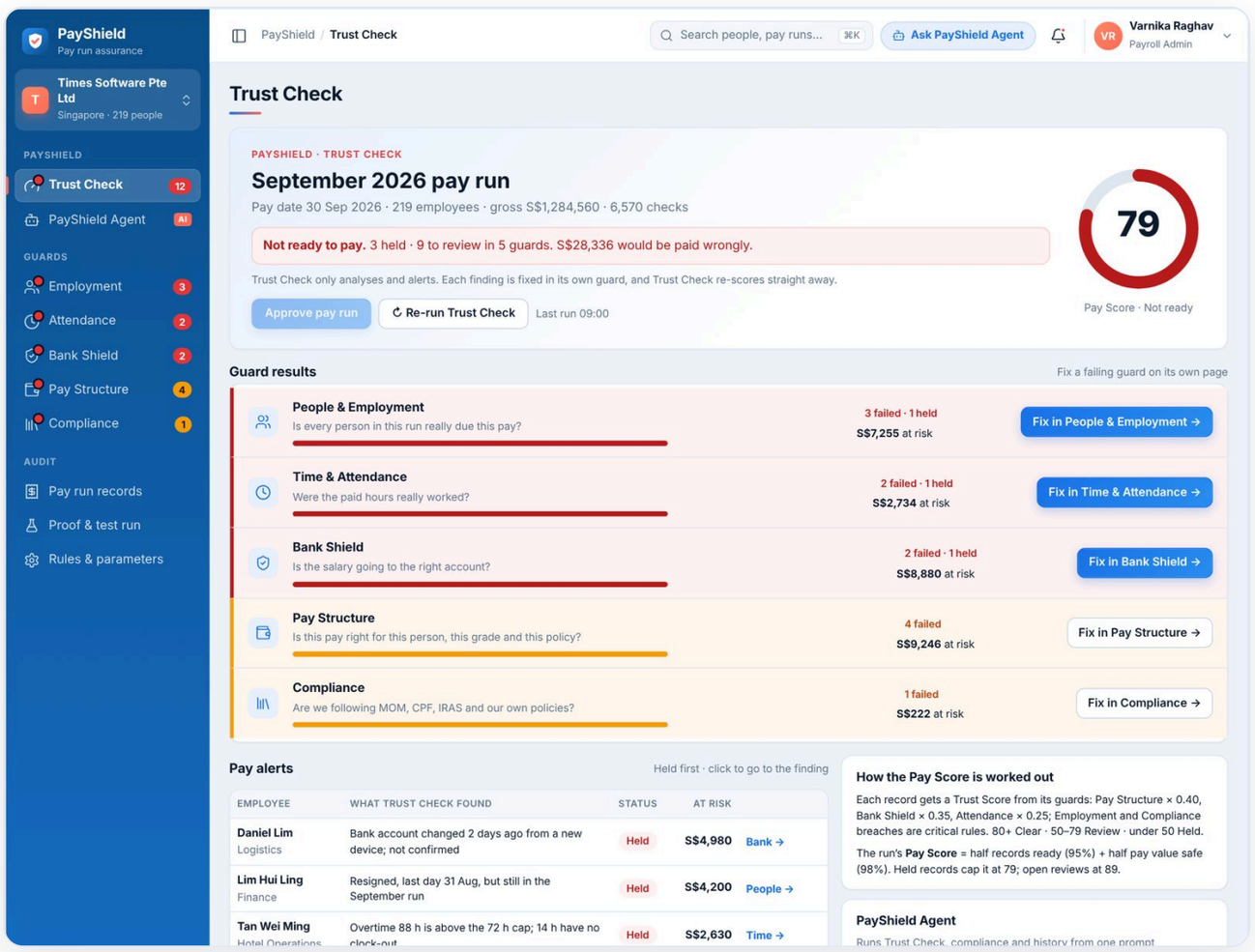
Why Phase 1 first

It uses only data already in Times Portal, needs no consent changes, and reduces reruns from the first pay run. Sprint estimates are early assumptions for the engineering team to confirm after reviewing the backend.

LIVE DEMO

See PayShield working end to end

The prototype opens on Trust Check. Click an alert to jump to the guard where it belongs, fix or accept it there, and watch the Pay Score reach 100 before Approve unlocks. ▶ buttons run step-by-step stories.



Trust Check: Pay Score, guard results and pay alerts

How to open it

Double-click **PayShield_Demo.html** in any browser, or run `npm install` and `npm run dev`. 57 automated tests: `npm test`.

Product pages (click freely)

PayShield: Trust Check, PayShield Agent · Guards: Employment, Attendance, Bank Shield, Pay Structure, Compliance · Audit: Pay run records, Proof, Rules & parameters

Trust Check only analyses and alerts. Fixes happen on the guard pages: findings with Fix / Accept, live validators and parameters, a red dot while anything is pending.

▶ Demo stories

1. **PayShield Agent: check the whole run** · One prompt → Pay Score → decide each finding (saved to PayShield) → approver → pay run recorded
2. **Bank-change attack** · Stolen login changes the account; two endings
3. **Fraud Network** · Every company protects the others
4. **Buddy punching** · Geofence, live selfie, missed clock-out
5. **Rule Watch** · AI drafts a CPF change; a person publishes
6. **Our own model** · Train it live, test on 20 new companies
7. **Test run** · 50 synthetic companies, 25 hidden problems each

Suggested 3-minute pitch path

Trust Check (Pay Score 79) → click Daniel's alert → fix in Bank Shield → score re-calculates → PayShield Agent: "check the pay run" → Pay Score 79 → decide each finding A/B/C (saved to the guards) → Pay Score 100 → approver → agent runs the pay run → audit record.